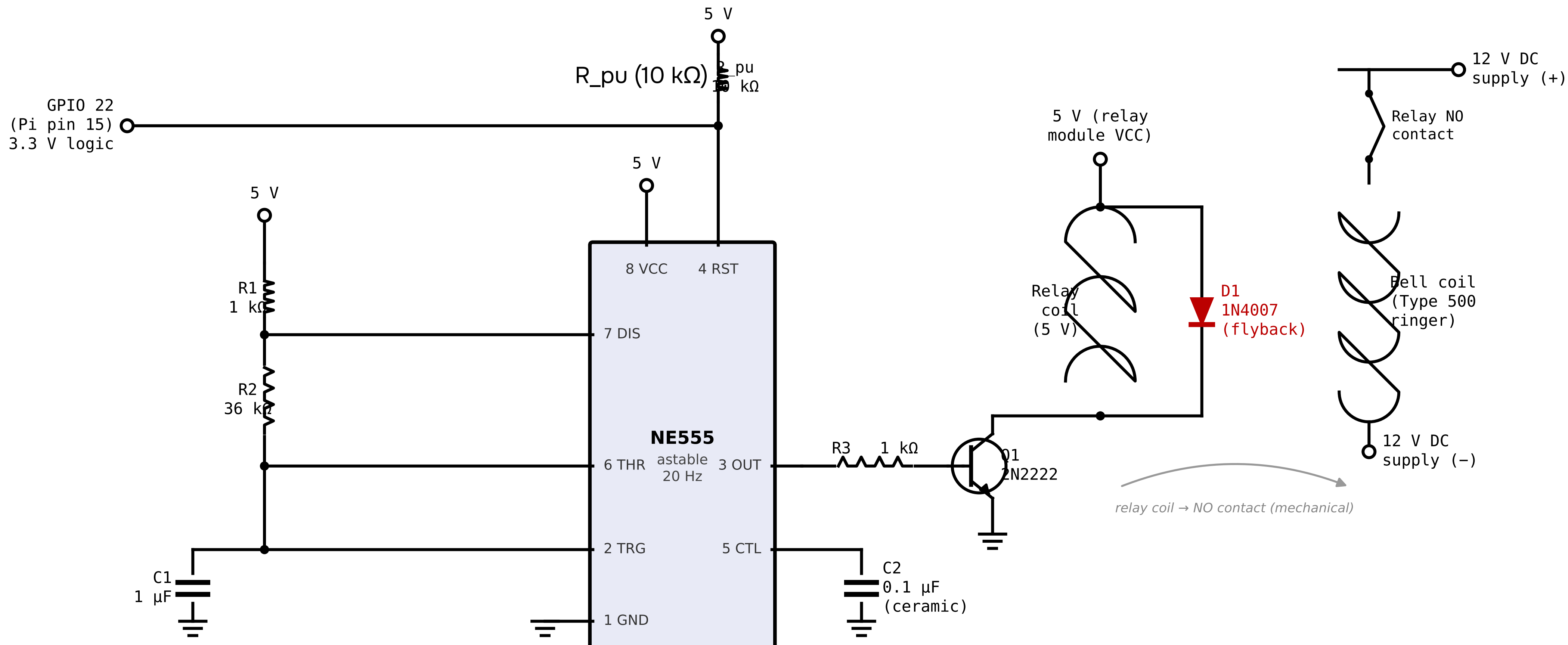
Bell Wiring — Option 2: 555 Astable at 20 Hz, Gated by GPIO 22
Authentic mechanical ring — clapper oscillates at 20 Hz during ring cadence




```
555 astable frequency:  f = 1.44 ÷ ((R1 + 2×R2) × C1)  =  1.44 ÷ (73 000 × 0.000001)  ≈  19.7 Hz  (nominal 20 Hz)
GPIO 22 HIGH → RESET (pin 4) released → 555 oscillates → 20 Hz square wave on OUT → Q1 switches relay at 20 Hz → clapper oscillates → authentic bell ring.
GPIO 22 LOW  → RESET held LOW → 555 output frozen LOW → relay open → bell silent.  Ring cadence (2 s on / 4 s off) is still controlled by the script.
GPIO 22 is 3.3 V logic; 555 RESET LOW threshold ≈ 0.7 V for 5 V supply — direct connection works, no level-shifter needed.
R_pu (10 kΩ): pull-up from 5 V to RESET — keeps RESET HIGH if GPIO floats at boot, preventing spurious ringing during startup.
Q1 (2N2222 NPN): R3 = 1 kΩ base resistor.  Handles relay coil current ≤ 100 mA.  Substitute 2N3904 for coils ≤ 50 mA.
D1 (red) — flyback diode: ESSENTIAL across relay coil; cathode (banded end) toward 5 V rail.  Without it Q1 fails instantly on de-energise.
C2 (0.1 µF ceramic): bypass cap on 555 pin 5 (CTL) — suppresses VCC noise coupling into timing; place as close to IC as possible.
Tune frequency: R2 = 27 kΩ → ≈25 Hz;  R2 = 47 kΩ → ≈15 Hz.  Bell electromagnet is rated 20 Hz; stay within 16–25 Hz for reliable ringing.
```